

Wine Quality Data Workflow: Module Summary

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Overview

I built a complete and reproducible Python data workflow that ingests, cleans, explores and visualises the Wine Quality dataset from the UCI Machine Learning Repository (Cortez et al., 2009a): <https://archive.ics.uci.edu/dataset/186/wine+quality>. The workflow is a Jupyter notebook (*data_workflow.ipynb*) supported by a README, a *requirements.txt* file and a GitHub repository with multiple commits and a development branch. No machine learning models are trained; the purpose is to create a clean foundation for later AI projects.

Dataset Description

The dataset was created for the study by Cortez et al. (2009b) and describes red and white variants of the Portuguese Vinho Verde wine. Each row is one wine sample with 11 physicochemical measurements (for example alcohol, density, volatile acidity and chlorides) and a sensory quality score, which the dataset documentation describes as the median of at least three evaluations by wine experts on a 0 to 10 scale (Cortez et al., 2009a). The raw files contain 6,497 rows (1,599 red and 4,898 white) and 12 columns, and I added a thirteenth column, *wine_type*. After cleaning, 5,320 rows remain (1,359 red and 3,961 white). My analysis focused on quality, alcohol, density, volatile acidity, chlorides and the relationships among the chemical measurements.

Workflow Description

Ingestion. A function reads the two semicolon-separated files and combines them with a wine type label. **Cleaning.** Four documented functions standardise column names, check data quality, remove exact duplicate rows and derive an ordered quality band (low, medium, high). **Exploratory analysis.** Three functions produce descriptive statistics, grouped summaries and a ranking of correlations with quality. **Visualizations.** Four labelled figures show quality by wine type, alcohol by quality, a correlation matrix and the alcohol and density relationship. **Summary.** The notebook ends with a written interpretation of patterns, limitations and open questions.

Key Decisions and Assumptions

Workflow and reproducibility decisions. I chose a Jupyter notebook because notebooks combine code, documentation, visualisations and narrative in one executable document, which is the approach taken by the open learning resource on reproducible data science with Python (Danchev, 2022). Following the same resource, I recorded package versions in a *requirements.txt* file so that others can recreate the environment. I kept all cleaning in scripted functions rather than editing the files by hand, because manual data manipulation is error-prone and hard to reproduce (Sandve et al., 2013), and I left the two raw CSV files unchanged so there is one authoritative copy of the raw data (Wilson et al., 2014). The code is organised into small functions with docstrings, in line with recommendations to write modular code and document its purpose (Wilson et al., 2014). Work was tracked with Git, using commits and a development branch, because version control records how the code evolved (Sandve et al., 2013; Wilson et al., 2014).

Cleaning choices. The checks found no missing cells, no negative values and no impossible pH or quality values, so no imputation was needed. The only substantive issue was 1,177 exact duplicate rows (240 red and 937 white), which I removed so that no measurement is counted twice. I assumed these were redundant records; because two different wines could share identical values, I checked that the mean quality of each wine type changed by less than 0.03 points after removal. Outliers were retained because extreme values are chemically possible and I had no domain basis for deleting them.

EDA focus and plot design. I focused on what is associated with quality, because that is the natural target for later modelling. Spearman correlation was reported alongside Pearson because quality is an ordered score. Figure 1 compares the quality profile of red and white wines as proportions so that unequal sample sizes do not distort the comparison. Figure 2 shows how alcohol, the strongest correlate, varies across quality scores. Figure 3 shows all pairwise correlations to reveal relationships among predictors. Figure 4 shows the alcohol and density relationship by quality band. The plots were produced with Seaborn, a library for statistical graphics built on Matplotlib (Waskom, 2021).

Results and Interpretation

Alcohol is the clearest signal in the data. It has the strongest correlation with quality (Pearson 0.469, Spearman 0.480), and median alcohol rises from about 9.6 percent for wines scoring 5 to 12.2 percent for wines scoring 8 (Figure 2). Density (Spearman -0.349), chlorides (-0.304) and volatile acidity (-0.251) are negatively related to quality. White wines are rated somewhat higher than red wines: 20.8 percent of white wines reach the high band against 13.5 percent of red wines, while 47.1 percent of red wines fall in the low band against 34.0 percent of white wines (Figure 1). The chemical variables are also strongly related to each other, for example alcohol and density (-0.67) and free and total sulfur dioxide (0.72) (Figure 3, Figure 4). No feature has a correlation with quality above 0.5, so no single measurement explains quality on its own.

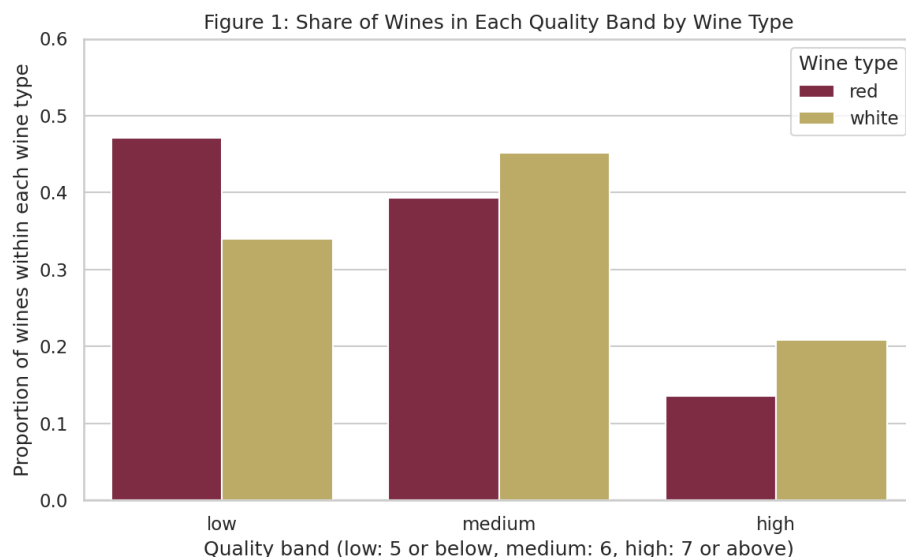


Figure 1. Share of wines in each quality band by wine type.

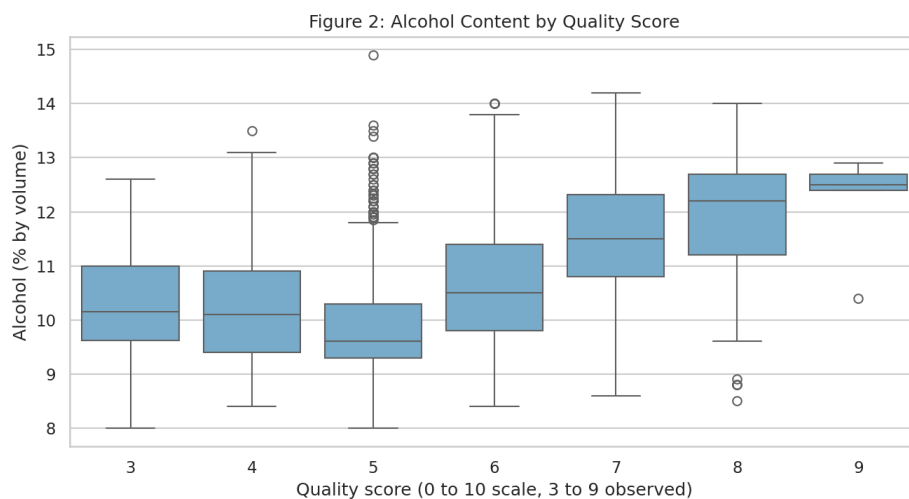


Figure 2. Alcohol content by quality score.

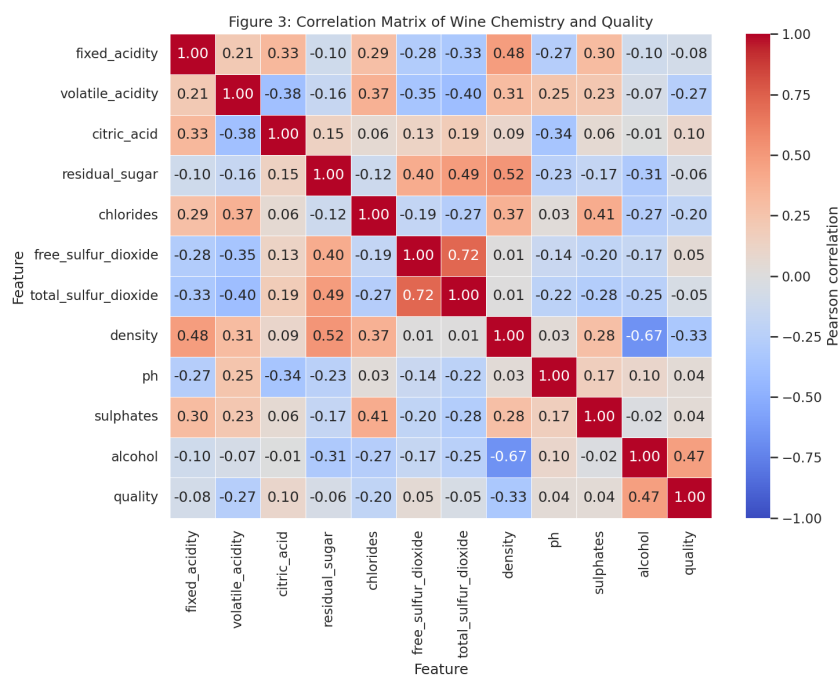


Figure 3. Correlation matrix of wine chemistry and quality.

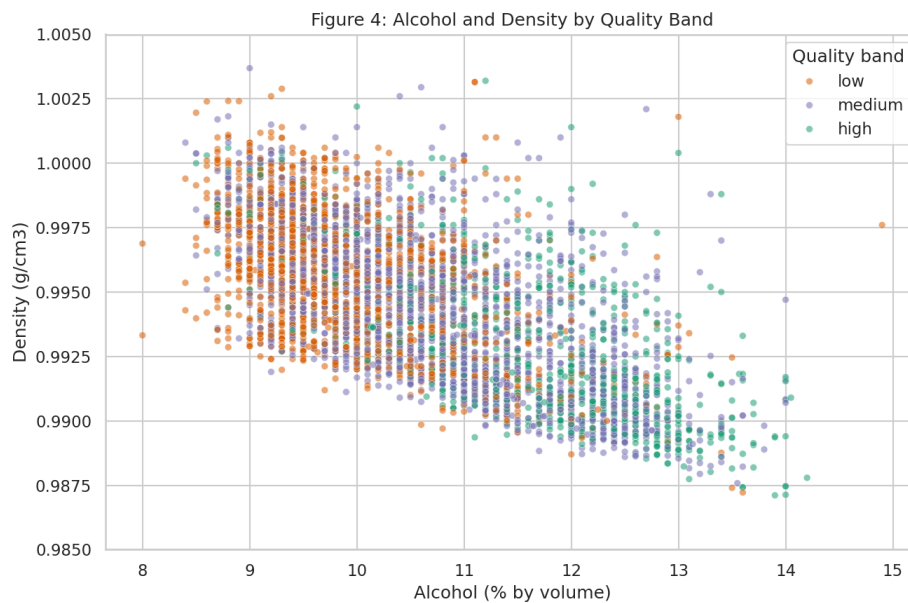


Figure 4. Alcohol and density by quality band.

Responsible Practice (Bias and Data Quality)

Several data handling choices could mislead. Removing duplicates affected white wines more (19.1 percent of rows) than red wines (15.0 percent), so the balance between types shifted; I mitigated this by reporting the uneven removal and by confirming that mean quality was almost unchanged, but I cannot be sure that every removed row was a true duplicate. Deleting outliers would have made the data look more typical than it is, so they were kept. Grouping scores into bands hides detail, particularly at the extremes where there are very few wines (30 wines scored 3 and only 5 scored 9). The data also has limited coverage: it covers one Portuguese region, the white wines outnumber red by about three to one, and the quality score is a subjective panel rating rather than a physical measurement (Cortez et al., 2009a). Conclusions should therefore not be generalised to other wines. Next steps would be to ask the data owners whether duplicates are genuine repeated samples, to compare results separately for each wine type, and to document these checks alongside any later model.

Reproducibility

Anyone can rerun the work by cloning the repository (<https://github.com/dohigg1987/ai-programming-foundations-project>), installing the packages with `pip install -r requirements.txt` and running `data_workflow.ipynb` with Restart and Run All. The `requirements.txt` file was created with `pip freeze` and pins the package versions used, which follows the practice demonstrated by Danchev (2022). The two raw CSV files are stored unchanged in the repository, and all cleaning happens in code, so every result can be traced back to the raw data (Sandve et al., 2013). The Git history shows the project built in stages across multiple commits, with development work on a separate branch before merging to the main branch (Wilson et al., 2014).

References

- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009a). *Wine quality* [Data set]. UCI Machine Learning Repository. <https://doi.org/10.24432/C56S3T>
- Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009b). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, 47(4), 547-553.

<https://doi.org/10.1016/j.dss.2009.05.016>

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), Article 156. <https://doi.org/10.21105/jose.00156>

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLOS Computational Biology*, 9(10), Article e1003285. <https://doi.org/10.1371/journal.pcbi.1003285>

Waskom, M. L. (2021). seaborn: Statistical data visualization. *Journal of Open Source Software*, 6(60), Article 3021. <https://doi.org/10.21105/joss.03021>

Wilson, G., Aruliah, D. A., Brown, C. T., Chue Hong, N. P., Davis, M., Guy, R. T., Haddock, S. H. D., Huff, K. D., Mitchell, I. M., Plumbley, M. D., Waugh, B., White, E. P., & Wilson, P. (2014). Best practices for scientific computing. *PLOS Biology*, 12(1), Article e1001745. <https://doi.org/10.1371/journal.pbio.1001745>